

Design and Finite Element Analysis of a 2-Finger Parallel Jaw Industrial Gripper

Industrial Design Mini Project Report

Submitted by
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1. Introduction

Robotic automation in manufacturing depends heavily on reliable end-of-arm tooling (EOAT). Among the various gripper types used in industry, parallel jaw designs remain one of the most widely adopted solutions for pick-and-place tasks involving objects with flat or prismatic surfaces.

This report documents the mechanical design and structural validation of a 2-finger parallel jaw gripper built to handle payloads between 5 kg and 10 kg. A gear-driven dual parallelogram linkage converts rotational actuation into synchronised parallel jaw motion. The gripper was modelled parametrically in SolidWorks and its structural performance was verified through static finite element analysis in ANSYS Workbench 2026 R1.

A factor of safety no lower than 1.5 under peak loading was set as the design target. The report covers the working principle, CAD modelling, material choices, force calculations, simulation results, and practical aspects such as manufacturing, assembly, and potential failure modes.

2. Working Principle of the Gripper

The mechanism uses a gear-driven dual parallelogram linkage to convert rotational input into synchronised, parallel linear motion of two gripping jaws. Both jaws move symmetrically and stay parallel throughout the full stroke.

2.1 Input Actuation and Power Transmission

A rotary actuator (servo motor or rotary pneumatic cylinder) drives the primary gear ('gear drive') through a central shaft. The front view in Figure 4 shows the central bore that accepts this input shaft. The driving gear meshes with a secondary driven gear ('gear drive 2') of identical module and pitch. Because they mesh directly, the two gears rotate in opposite directions at equal angular velocity. This contra-rotation is what produces the symmetric jaw motion.

2.2 Motion Conversion and Parallel Jaw Kinematics

Each gear body has an extended crank arm formed as an integral feature. The tip of each crank arm connects to the corresponding gripping element (finger) through a revolute pin joint. A single crank alone would cause the finger to rotate as it moves, changing its orientation relative to the payload. To prevent this, a secondary link called the 'aligner' is used for each finger. One end of the aligner is pinned to the stationary housing; the other is pinned to the gripping element. Because the pivot-to-pivot distance on the housing matches that on the gripping element, the result is a classic four-bar parallelogram linkage for each jaw.

2.3 Synchronisation and Load Path

The 1:1 gear ratio ensures both linkages actuate at the same rate. During gripping, the clamping force passes from the jaw contact surfaces through the rigid finger body into the revolute pins, then through the crank arms (which see bending moments on the gear bodies) and the aligner links (acting mainly as

two-force members in tension or compression). All reaction forces ultimately anchor into the stationary housing, which transfers the payload weight to the robotic arm.

3. CAD Model and Assembly

The gripper was modelled parametrically in SolidWorks so that key dimensions can be updated without rebuilding the geometry from scratch. The assembly comprises a housing, a housing cover, two gear drives with integrated crank arms, two aligner links, two gripping elements with serrated pads, spacers in three lengths (15 mm, 20 mm, 30 mm), and standard hex fasteners.

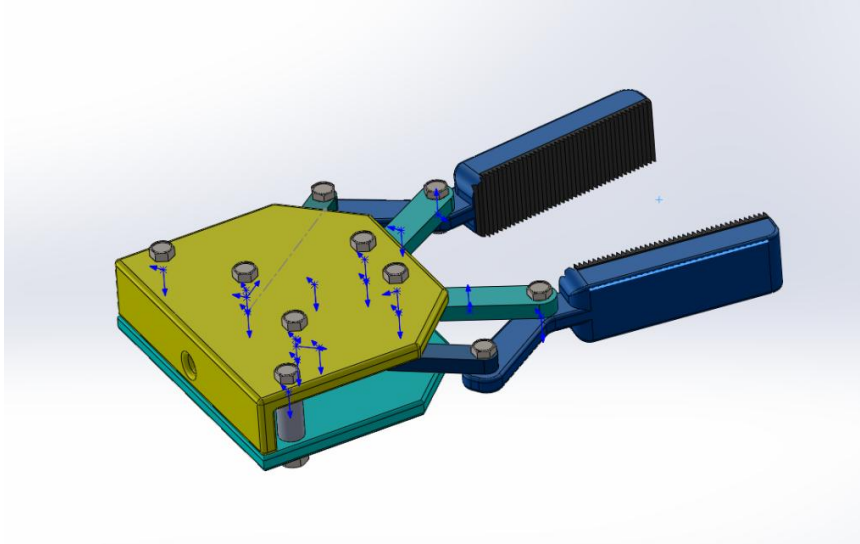


Figure 1: Isometric view of the assembled 2-finger parallel jaw gripper

Figure 1 shows the complete assembly in isometric view. The yellow housing cover encloses the gear mechanism, while the teal housing base provides the structural foundation. Two dark blue gripping elements extend

outward, connected through the teal aligner links and gear-driven crank arms. The serrated gripping pads (black) on the inner faces of the fingers can be seen at the distal ends.

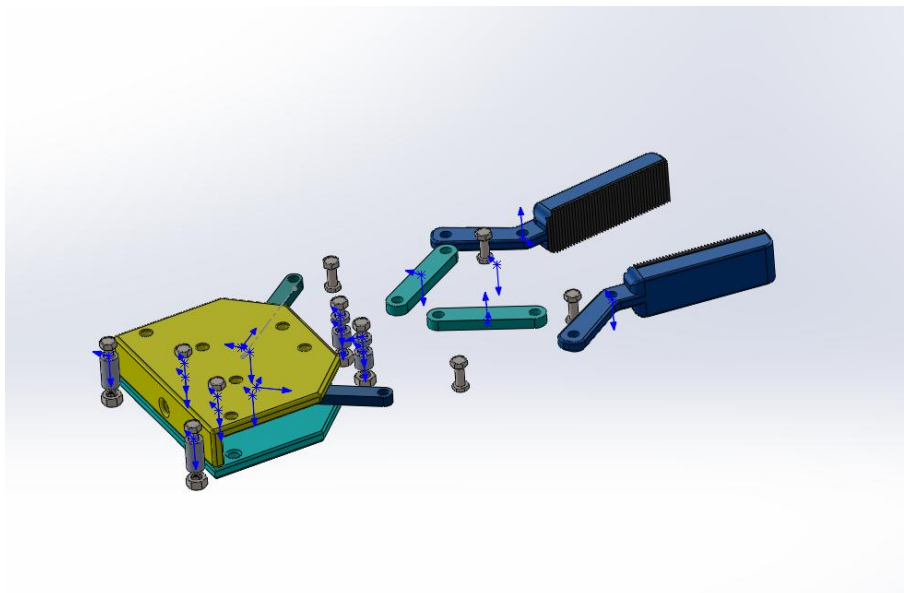


Figure 2: Exploded view showing all individual components of the gripper assembly

The exploded view in Figure 2 reveals the internal arrangement. On the left, the housing base shows the machined pivot bores. The two flat aligner links (teal) and the gear-crank assemblies connect to the gripping elements on the right. Multiple spacers sit between the stacked components to maintain axial alignment, and hex-head fasteners clamp everything together.

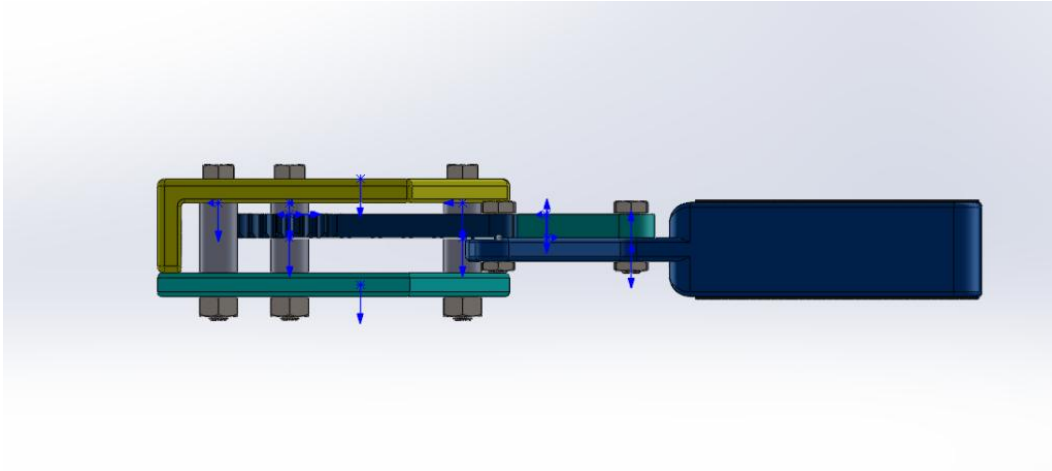


Figure 3: Side view of the gripper assembly showing the linkage arrangement

The side elevation in Figure 3 highlights the layered construction: the housing base and cover sandwich the internal gears and spacers, while the aligner and crank arm extend toward the gripping element. The compact vertical profile of the design is apparent from this view.

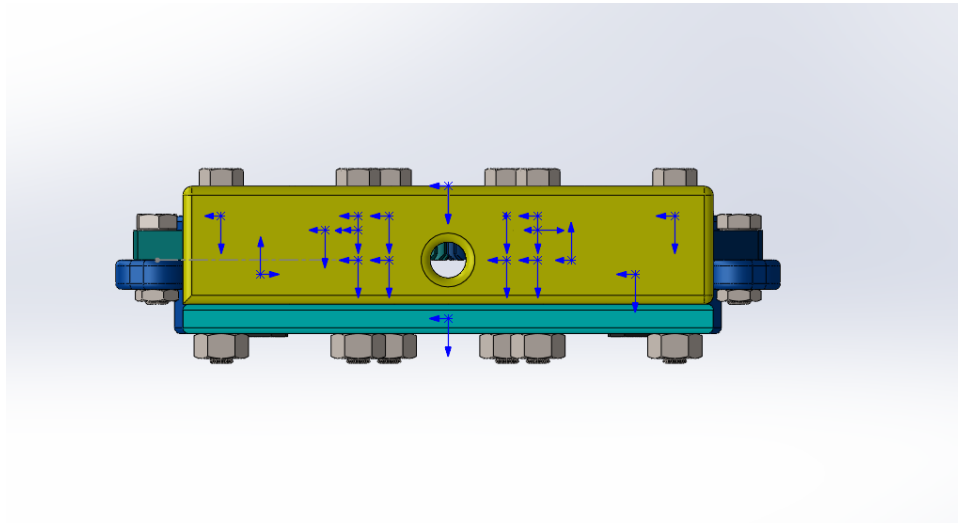


Figure 4: Front view of the gripper showing the central actuator bore and symmetric layout

Figure 4 presents the front view. The central bore on the housing cover is the access point for the rotary actuator input shaft. Pivot bores on both sides of the cover and the fastener locations are visible. The two gripping elements protrude symmetrically on either side.

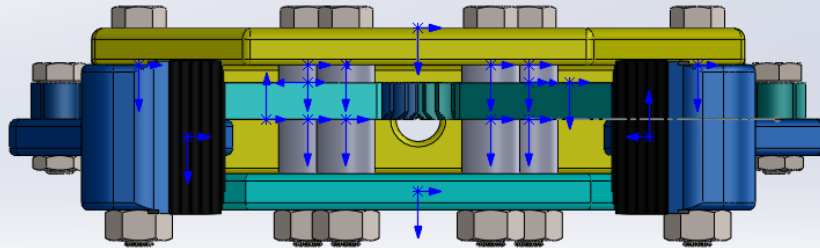


Figure 5: Top view of the gripper showing internal gear arrangement and finger symmetry

The plan view in Figure 5 is useful for checking the symmetric arrangement of the two parallelogram linkages. The meshing gears are partially visible between the cover and base. Both gripping elements and their serrated pads extend outward.

4. Component Descriptions

4.1 Housing and Housing Cover

The housing is the structural backbone of the gripper. It contains precision-machined bores for the gear shafts, aligner pivots, and crank arm pivots, and it provides the mounting interface to the robotic arm. The housing cover encloses the gear mechanism from above, shielding internals from particulate contamination and placing the pivot pins in double-shear to stiffen the joints. Both parts are aluminium to keep the EOAT mass low.

4.2 Driving Gear and Driven Gear

The two gears (gear drive and gear drive 2) share the same module and tooth count. Each gear body includes an extended crank arm that transmits rotational motion to the gripping element through a pin joint. Structural steel was chosen for the gears because of the Hertzian contact stresses at the meshing flanks and the bending stresses on the crank arms.

4.3 Aligner Links

Each aligner is a simple two-hole flat link pinned between the housing and the gripping element, forming the second bar of the parallelogram. Its job is to prevent the finger from rotating during actuation. Under load, the aligners act mainly as two-force members in tension or compression.

4.4 Gripping Elements

These are the fingers. The internal contact faces carry serrated or grooved pads to raise the friction coefficient against the workpiece. Each finger has two pin bores: one connects to the gear crank arm, the other to the aligner. They are structural steel to resist bending under the clamping loads.

4.5 Spacers and Fasteners

Cylindrical spacers in three lengths (15 mm, 20 mm, 30 mm) maintain axial separation between the housing, cover, gears, and linkage components, preventing interference during actuation. Standard hex-head fasteners (bolts and nuts) clamp the sandwich together, placing the pins in double-shear.

5. Material Selection and Justification

Two materials were used in the ANSYS simulation. Their properties are listed in Table 1.

Property	Structural Steel	Aluminium
Density (kg/m ³)	7850	2710
Young's Modulus (Pa)	2.0×10^{11}	6.89×10^{10}
Poisson's Ratio	0.3	0.33
Tensile Yield Strength (Pa)	2.5×10^8	2.9×10^8

Table 1: Material properties assigned in ANSYS simulation

5.1 Aluminium (Housing and Cover)

The housing and cover were assigned aluminium (density 2710 kg/m³). A heavy gripper eats into the robot's payload budget and limits how fast the arm can accelerate, so keeping the static structural bodies light was a priority. With a yield strength of 290 MPa and a Young's modulus of 68.9 GPa, aluminium comfortably handles the bearing loads from the pivot pins. It also machines well and resists corrosion without special coatings.

5.2 Structural Steel (Gears, Aligners, Gripping Elements, Spacers, Fasteners)

All moving components are structural steel (yield strength 250 MPa, Young's modulus 200 GPa, density 7850 kg/m³). The gears need it because the tooth flanks experience Hertzian contact pressures and cyclic bending at the root. The aligner links need the stiffness to avoid elastic stretching or buckling, which would ruin the parallel kinematics. The gripping elements need it to resist bending deflection under the clamping forces. Steel is heavier than aluminium, but its wear resistance, fatigue life, and stiffness are non-negotiable for these parts.

6. Design Calculations

6.1 Payload Weight

The maximum design payload is $m = 10$ kg. The weight of the payload is given by Equation (1):

$$W = m \times g \quad (1)$$

Substituting $m = 10 \text{ kg}$ and $g = 9.81 \text{ m/s}^2$:

$$W = 10 \times 9.81 = 98.1 \text{ N} \quad (2)$$

6.2 Required Holding Force with Safety Factor

To account for dynamic effects during robot motion, the project specification requires that the gripping system maintain at least a 1.5 times margin over the static payload weight. The total required holding force is therefore obtained from Equation (3):

$$F_{\text{hold}} = W \times SF = 98.1 \times 1.5 = 147.15 \text{ N} \quad (3)$$

6.3 Gripping Force per Finger

The payload is supported solely by static friction at the two jaw-workpiece interfaces. Taking the coefficient of static friction as $\mu = 0.6$ (a conservative value for steel on metal) and $n = 2$ fingers, vertical equilibrium gives Equation (4):

$$n \times \mu \times F_g = F_{\text{hold}} \quad (4)$$

Rearranging for the normal gripping force per finger, Equation (5):

$$F_g = F_{\text{hold}} / (n \times \mu) \quad (5)$$

Substituting the values:

$$F_g = 147.15 / (2 \times 0.6) = 122.6 \text{ N} \quad (6)$$

So each jaw needs to press against the workpiece with roughly 122.6 N to hold the full 10 kg payload.

6.4 Actuator Torque Estimation

Note: This is a first-order theoretical estimation assuming perpendicular force application and neglecting instantaneous linkage kinematics.

If the effective perpendicular lever arm on the driving gear crank is taken as approximately $L_{\text{arm}} = 50 \text{ mm}$ (0.05 m), the torque needed at the gear pivot follows from Equations (7) and (8):

$$T_{\text{req}} = F_g \times L_{\text{arm}} \quad (7)$$

$$T_{\text{req}} = 122.6 \times 0.05 = 6.13 \text{ Nm} \quad (8)$$

Both fingers are driven through the meshing gear pair, so the total input torque at the primary shaft is roughly 12.26 Nm plus whatever is lost to friction at the pin joints. A proper kinematic analysis of the instantaneous linkage geometry would refine this number.

7. Finite Element Analysis

Static structural analysis was carried out in ANSYS Workbench 2026 R1 (Student Edition) to check that the gripper can handle the calculated operating loads without yielding or deforming excessively.

7.1 Simulation Setup

The full gripper assembly was imported into ANSYS as a multi-body part. Materials were assigned per Table 1. Contacts between mating surfaces were set to bonded, which is conservative for a static study: it prevents relative sliding and transmits full bending moments across the interfaces.

7.2 Mesh Generation

An adaptive mesh was generated with 78,446 nodes and 39,515 elements. The algorithm automatically placed finer elements around pin bores, fillet radii, and cross-section transitions where stress gradients tend to be steep. Table 2 summarises the mesh details.

Parameter	Value
Nodes	78,446
Elements	39,515
Element Type	Tetrahedral (adaptive)
Contact Type	Bonded

Table 2: Mesh statistics

7.3 Boundary Conditions and Loads

Three boundary conditions were applied to mimic the physical mounting and loading, as summarised in Table 3.

Fixed Support: The rear mounting face of the housing was fully constrained (all six degrees of freedom set to zero), simulating the bolted connection to the robotic wrist.

Force 2 (Actuation Load): A ramped force of 61.3 N distributed across 18 edges of the gear drive and linkage components, representing the actuation loads transmitted through the mechanism.

Force 3 (Gripping Load): A ramped force of 150 N applied to the two contact faces of the gripping elements, representing the payload reaction during maximum clamping.

Load	Type	Magnitude	Application
Fixed Support	Constraint	-	Housing base face
Force 2	Ramped Force	61.3 N	18 edges (gear/linkage)
Force 3	Ramped Force	150 N	2 faces (gripping elements)

Table 3: Boundary conditions and applied loads

7.4 Total Deformation

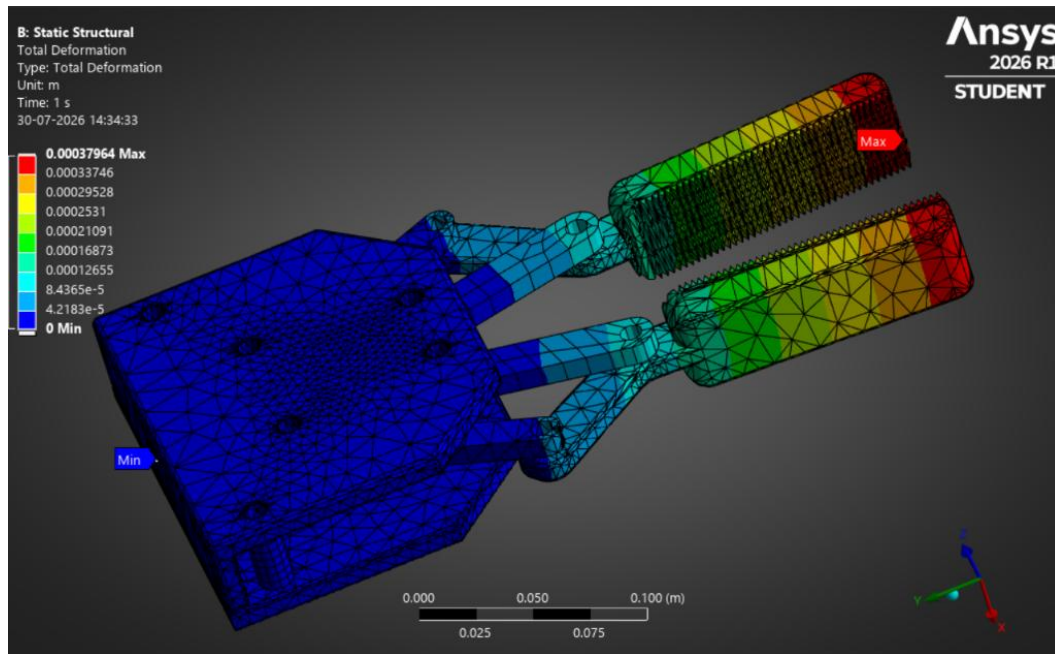


Figure 6: Total deformation contour plot (ANSYS 2026 R1). Maximum: $3.796 \times 10^{-4} \text{ m}$

Figure 6 maps the total deformation across the gripper. The peak value of $3.796 \times 10^{-4} \text{ m}$ (0.38 mm) appears at the distal tips of the gripping elements, marked by the red contour.

The gripping elements act as cantilever beams projecting from the pivot pins. The normal force at the tips creates a bending moment that grows along the finger length, producing maximum deflection at the free end. The housing, locked in place by the fixed support, sits at near-zero deformation (dark blue).

A deflection of 0.38 mm is unlikely to cause kinematic binding or compromise the grip during normal operation.

7.5 Equivalent (von-Mises) Stress

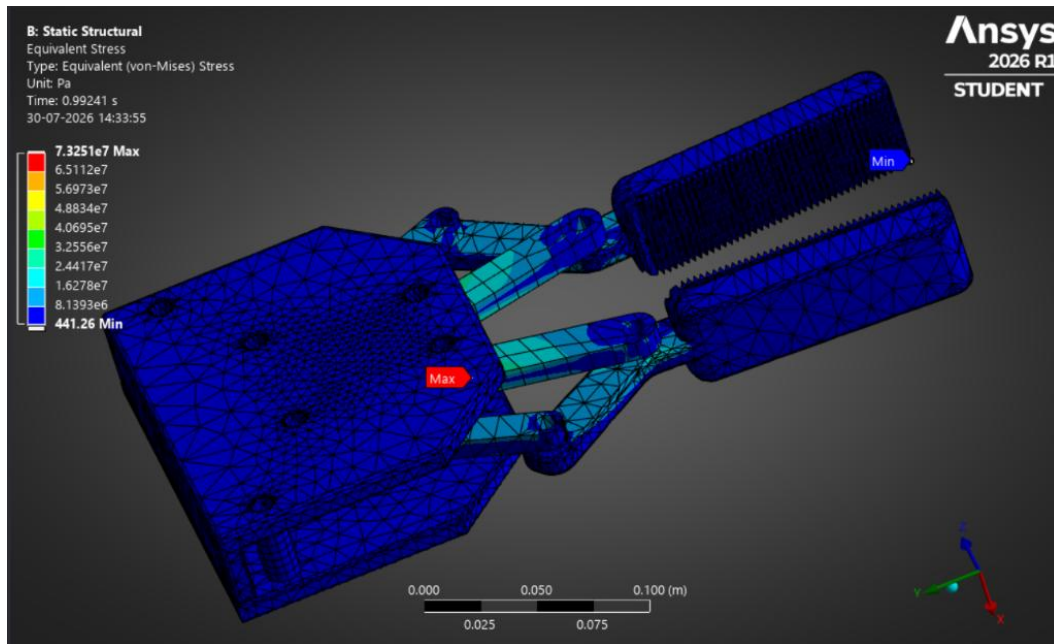


Figure 7: Equivalent (von-Mises) stress contour plot. Maximum: 7.3251×10^7 Pa (73.25 MPa)

The von-Mises stress distribution appears in Figure 7. Peak stress reaches 7.3251×10^7 Pa (73.25 MPa) at the base of the linkage arms near the pin joints, flagged by the 'Max' marker. The minimum value (441.26 Pa) is at the finger tips, tagged 'Min'.

This concentration occurs because the narrow linkage arms meet the wider gear body at a sharp cross-section change, which disrupts stress flow and creates a local peak. The bending moment from the gripping force is also highest at the arm root, where the internal lever arm is longest.

Most of the housing and the outer portions of the fingers carry very little stress. At 73.25 MPa, the peak is roughly 29% of the 250 MPa yield strength of structural steel.

7.6 Equivalent Elastic Strain

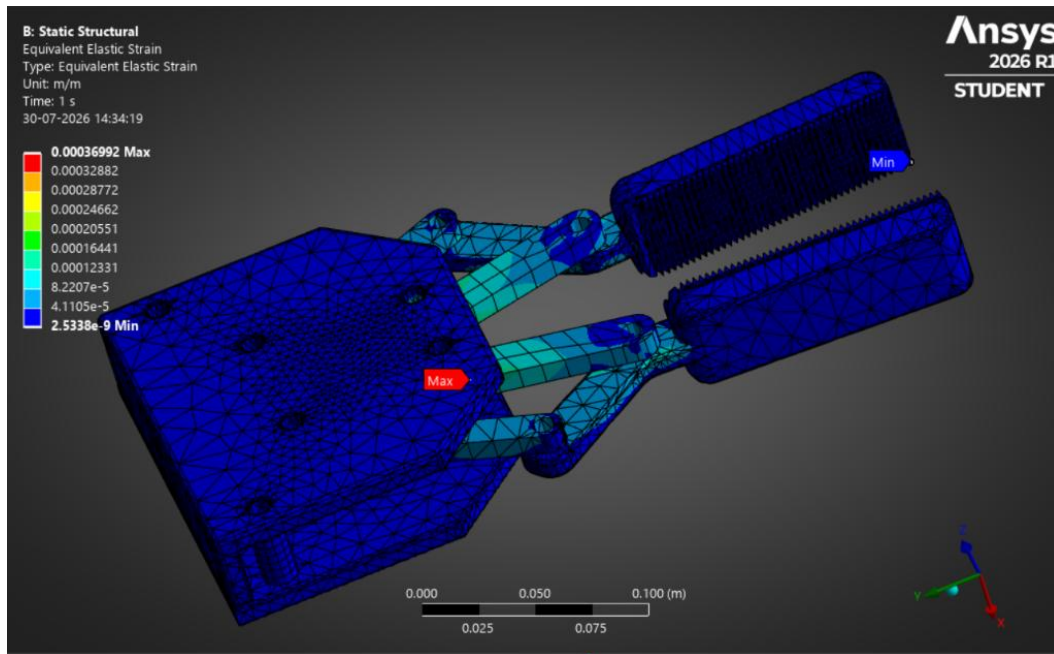


Figure 8: Equivalent elastic strain contour plot. Maximum: 3.6992×10^{-4} m/m

Figure 8 shows the elastic strain field. The maximum of 3.6992×10^{-4} m/m occurs at the same location as the peak stress (linkage arm base). The contour pattern mirrors the stress plot closely.

7.7 Factor of Safety

Since the highest stress sits in a structural steel component (yield strength 250 MPa), the minimum factor of safety follows from Equation (9):

$$FoS = \text{Yield Strength} / \text{Max. von-Mises Stress} \quad (9)$$

$$FoS = 250 / 73.25 = 3.41 \quad (10)$$

Result	Maximum Value	Location
Total Deformation	3.796×10^{-4} m (0.38 mm)	Gripping element tips
Von-Mises Stress	7.325×10^7 Pa (73.25 MPa)	Linkage arm base / pin joints
Elastic Strain	3.699×10^{-4} m/m	Linkage arm base / pin joints
Factor of Safety	3.41	Linkage arm base (minimum)

Table 4: Summary of FEA results

At 3.41, the FoS more than doubles the project requirement of 1.5. The design is over-engineered by that metric, which is partly deliberate (we wanted conservative first margins) and partly a consequence of using standard plate thicknesses rather than optimised sections. Topology optimisation on the housing and finger bodies could strip mass from the low-stress blue zones without compromising strength, and increasing fillet radii at the linkage junctions would spread the peak stress further.

8. Discussion and Mechanism Justification

We considered several alternative mechanisms before settling on the gear-driven dual parallelogram linkage.

A rack-and-pinion gives truly linear jaw motion, but it needs precision guide rails. Those rails add sliding friction, clog easily with swarf and dust, and make the housing wider. The parallelogram linkage replaces all that sliding with rolling/pivoting at pin joints, which is simpler to maintain and lasts longer in a dirty shop environment.

Scissor linkages pack a large stroke into a small space, but their mechanical advantage is highly non-linear, making consistent clamping across the stroke hard to guarantee.

Cam mechanisms were ruled out due to the high contact stress and rapid wear on the follower.

A pair of direct linear actuators (pneumatic cylinders) would be straightforward, but they lack inherent synchronisation between the two jaws.

9. Design Assumptions

We made several simplifying assumptions that should be kept in mind when interpreting the results:

1. The payload is rigid and incompressible with flat parallel grasping surfaces.
2. The coefficient of static friction (0.6) remains constant and is not degraded by surface contamination.
3. All pinned joints have zero clearance. In reality, manufacturing tolerances will introduce a small amount of play and kinematic hysteresis.
4. Dynamic inertial loads during robot motion are assumed to be covered by the 1.5 safety factor.
5. Bonded contacts were used in the simulation; real joints may experience micro-slip under load.
6. Thermal effects were neglected.

10. Assembly Procedure

Assembly follows a bottom-up sequence:

1. Secure the housing base to a fixture.
2. Place the driving and driven gears onto their shaft bores with the teeth correctly engaged.
3. Fit the spacers (15 mm, 20 mm, 30 mm) over the corresponding pivot posts.
4. Connect the aligner links and gripping elements to the crank arms and housing pivots simultaneously, forming the closed kinematic loops.
5. Set the housing cover in place.
6. Insert and torque the hex-head fasteners, clamping the pins in double-shear between the housing and cover.

11. Manufacturing Process

The main processes involved are:

1. CNC Milling: For the housing, cover, and gripping elements, to achieve tight dimensional tolerances and flat mating surfaces.
2. Gear Hobbing: To cut the involute tooth profiles on both gears.
3. Laser or Waterjet Cutting: For the flat aligner links from steel plate stock.
4. CNC Turning: For the cylindrical spacers, held to precise length tolerances.

12. Maintenance

Routine maintenance items:

1. Greasing the gear teeth.
2. Inspecting pin joints for radial play.
3. Re-torquing hex fasteners.
4. Checking gripping pads for wear.

13. Failure Modes

Identified failure modes:

1. Gear tooth shearing on impact.
2. Pin bore elongation from wear.
3. Payload slip due to pad contamination.
4. Fatigue cracking at the linkage arm.

14. Design Limitations

There are several limitations to the current analysis:

1. Only a static structural analysis was run. Because vibration, impact, and inertial forces during fast robot motions were not simulated, the dynamic behaviour and potential resonance modes remain entirely uncharacterised. A full transient dynamic study would be required to validate the gripper for high-speed pick-and-place trajectories.
2. Manufacturing tolerances on pin bores and gear profiles were ignored.
3. Joint clearances were not modelled in the FEA.
4. Contact friction at pin joints was simplified using bonded formulations.
5. The jaws follow a slight arc rather than a perfectly straight line, meaning the robot controller must compensate for a small vertical displacement during clamping.

15. Future Improvements

If we were to take this design further, the key improvements would be:

1. Adding compliant rubber or polyurethane pads on the jaw faces to boost friction and conform to

irregular payload geometries, which would significantly expand the range of objects the gripper could reliably handle without slipping.

2. Embedding strain-gauge sensors in the aligner links for closed-loop force feedback.
3. Running topology optimisation on the housing to reduce mass.
4. Performing a fatigue analysis.
5. Conducting a dynamic simulation under realistic robot acceleration profiles.

16. Applications

This gripper is intended for:

1. Automated CNC machine tending (loading and unloading workpieces).
2. Packaging and palletising of medium-weight components (5 to 10 kg).
3. Automotive assembly line pick-and-place.
4. General industrial handling of payloads with flat, parallel grasping surfaces.

17. Cost Considerations

Using standard steel plate for the aligners and off-the-shelf hex fasteners keeps the BOM cost low. The main expense is in the custom-hobbed gears and CNC-milled housing. In volume production the aluminium housing could be die-cast instead of machined, which would cut the per-unit cost substantially. Overall the design avoids exotic materials and complex geometry, keeping prototype costs reasonable for a university project.

18. Conclusion

This project produced a 2-finger parallel jaw gripper driven by a gear-coupled dual parallelogram linkage. The mechanism delivers synchronised, parallel jaw motion from a single rotary input. Hand calculations showed that each finger must apply 122.6 N of normal force to securely hold a 10 kg payload, given a coefficient of friction of 0.6 and a 1.5 times safety margin on the payload weight.

ANSYS static structural analysis returned a peak von-Mises stress of 73.25 MPa at the linkage arm roots and a maximum tip deflection of 0.38 mm. Against the 250 MPa yield strength of structural steel, that gives a factor of safety of 3.41 — well above the 1.5 target. The gripper is structurally adequate for the intended payload range.

Possible next steps include dynamic simulation, fatigue analysis, topology optimisation for mass reduction, and integration of force feedback sensors.